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Connecting display screen to device body

Abstract

Examples are disclosed that relate to securing a display screen of a display device to a body of the display device. One example provides a display device including a body, a display screen, and a connecting structure comprising a tab component having one or more tabs, the tab component being coupled to one of the display screen and the body, a hook component including one or more hooks complementary to the one or more tabs, the hook component coupled to the other of the display screen and the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and an actuation mechanism accessible from an exterior side of the body, the actuation mechanism being actuatable to engage the tab component and the hook component.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/754,378 filed Mar. 31, 2022, which is a U.S. National Phase of International Patent Application Serial No. PCT/US2020/056438 entitled “CONNECTING DISPLAY SCREEN TO DEVICE BODY”, filed Oct. 20, 2020, which claims priority to Netherlands Patent Application Serial No. 2024096, filed Oct. 24, 2019, the entire contents of each of which are hereby incorporated by reference for all purposes.

BACKGROUND

(1) Display devices commonly include a display screen coupled to a device body. Display devices with thin profiles may utilize an adhesive (e.g. a pressure sensitive adhesive) or mechanical snap-fit components to couple a display screen to a device body. Such connections may help to avoid the use of bulky brackets and/or wide bezels.

SUMMARY

(2) Examples are disclosed that relate to removably securing a display screen to a body of a display device. One example provides a display device comprising a body, a display screen, and a connecting structure comprising a tab component having one or more tabs, the tab component being coupled to one of the display screen and the body, a hook component comprising one or more

hooks complementary to the one or more tabs, the hook component coupled to the other of the display screen and the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and an actuation mechanism accessible from an exterior side of the body, the actuation mechanism being actuatable to engage the tab component and the hook component.

(3) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-1B show an example display device in the form of a laptop computer.
- (2) FIG. 2 schematically illustrates an example connecting structure for securing a display screen of a display device to a body of the display device.
- (3) FIG. 3 illustrates a perspective view of the example connecting structure of FIG. 2.
- (4) FIG. 4 schematically illustrates another example connecting structure for securing a display screen to a body of a display device.
- (5) FIG. 5 schematically illustrates aspects of an example actuation mechanism comprising a rotatable linkage.
- (6) FIG. 6 schematically illustrates example regions of a display device for integrating buttons, ports, and/or other interface hardware.
- (7) FIG. 7 schematically illustrates an example frictional connector for securing a display screen of a display device to a body of the display device via a frictional press fit in a region comprising interface hardware.
- (8) FIG. 8 schematically illustrates an example fin for securing a display screen to a body of a display device.
- (9) FIG. 9 is a flowchart illustrating an example method of securing a display screen to a body of a display device.

DETAILED DESCRIPTION

(10) Display replacement is a common repair for display devices, both for devices dropped during use and devices damaged during manufacturing. Examples of display devices include laptop computing devices, tablet computing devices, mobile phones, televisions, and computer monitors. Various repair and/or diagnostic services also may involve removing a display screen to access internal hardware.

(11) As mentioned above, a display screen may be connected to a display device body via snap fit connections or adhesives, as examples. However, each of these connections poses various drawbacks. For example, snap-fit components may not retain a screen as securely as an adhesive, and/or may provide less structural rigidity and exhibit poor flatness. Adhesive connections may provide a more uniform distribution of the retention force compared to snap-fit components, but may be difficult to separate without damaging the display screen.

(12) Some display devices may be configured to be disassembled via a debonding tool that utilizes heating plates to heat edges of the display screen where the display screen is adhered to the device body. After softening the adhesive via heat, the debonding tool may use suction to pull the display screen away from the device body gently. While this may help to prevent damage to a display screen during removal, such a debonding process may be time-consuming, expensive, and require use of specialized tools.

(13) Accordingly, examples are disclosed that relate to securing a display screen to a display device body in a manner that may facilitate display screen removal. Briefly, the disclosed examples utilize a connecting structure comprising a tab component coupled to one of a display screen and device body, and a complementary hook component coupled to the other of the display screen and device body. The tab component and hook component are moveable relative to one another to selectively and reversibly couple the display screen to the device body via an actuation mechanism accessible from an exterior side of the device body. As described in more detail below, such a connection may be operated without using special equipment and may provide for the simple and repeatable removal of the display without damage from disassembly, thereby increasing serviceability compared to adhesives, as well as facilitating removal and re-installation of a display screen without generating waste materials. Further, such a connection may allow for a relatively thinner bezel than an adhesive connection.

(14) FIGS. 1A-1B show an example display device **100** in the form of a laptop computer. The display device **100** includes a keyboard portion **102** and a display portion **104** coupled to the keyboard portion **102** via a hinge **106**. The display portion **104** includes a display screen **108** and a body **110** that supports the display screen **108**. Other example display devices include tablet computers, mobile phones, computer monitors, televisions, and desktop all-in-one computers.

(15) FIG. 2 schematically shows a sectional view of the display portion **104** (taken along line 2-2 of FIG. 3), and illustrates a connecting structure **202** that couples the body **110** to the display screen **108**. The term “display screen” as used herein refers to an image-producing device **204** (e.g. an organic light emitting diode (OLED), liquid crystal, quantum dot light emitting diode (QLED), or other display panel) as well as a protective cover sheet **206** (e.g. glass). In some examples, the display screen **108** also may include a touch sensor configured to sense touch inputs to the display screen **108**.

(16) The depicted connecting structure **202** functions as a cinching mechanism that selectively and removably secures the display screen **108** to the body **110**. The connecting structure **202** includes a tab component **208** having one or more tabs **210**, a hook component **212** having one or more hooks **214** complementary to the one or more tabs **210**, and an actuation mechanism (not shown in FIG. 2) that is accessible from an exterior side of the body **110**. The tab component **208** and the hook component **212** are movable relative to one another to selectively engage or disengage the one or more tabs **210** with the one or more hooks **214**.

(17) In the example of FIG. 2, the tab component **208** is slidably positioned within a slot **216** formed in the body **110**, and the hook component **212** is fixed to a back surface of the protective cover sheet **206**. While the disclosed examples are described in the context of the tab component **208** being coupled to the body **110** and the hook component **212** being coupled to the display screen **108**, in other examples the tab component **208** may be coupled to the display screen **108** and the hook component **212** to the body **110**.

(18) FIG. 3 shows a perspective view of the display device **100** of FIG. 2, with the display screen removed to illustrate the connecting structure **202** components in more detail. In this figure, the slot **216** can be seen extending along an interior side of the body **110**. The slot **216** may have any suitable depth depending upon the thickness of the device body **110** and dimension of the tab component **208**. In some examples, the slot **216** comprises a depth in a range of 0.5 to 3.0 millimeters (mm). The tab component **208** is inserted in the slot **216** such that the one or more tabs **210** extend in a direction away from the slot. In some examples, the slot **216** may be formed along only a portion of the interior side of the body **110**, while in other examples the slot may extend along an entirety of the inner side of the body. Further, the display device **100** may include slots extending along other interior sides to accommodate other connecting structures for other device sides.

(19) The one or more hooks **214** and the one or more tabs **210** may be configured to provide a selected force between the display screen **108** and the body **110** based upon a particular device in

which the connecting structure is being used. Further example, the one or more hooks and/or the one or more tabs each may comprise a sloped surface configured to increase the force at which the display screen **108** is pulled against the body **110** as the hook component **212** and tab component **208** progressively engage.

(20) The tab component **208** and the hook component **212** may each be formed from any suitable material(s). In some examples, the tab component **208** and/or the hook component **212** each is formed from a stamped metal sheet. In other examples, the tab component **208** and/or the hook component **212** each is formed from a molded polymeric material having suitable hardness and wear resistance, such as polycarbonate, acrylonitrile butadiene styrene (ABS), polyethylene terephthalate (PET), or nylon. Example methods for forming the tab component **208** and/or the hook component **212** include stamping, molding, casting, and 3D printing, depending upon the material or material from which these parts are formed.

(21) FIG. 4 depicts another example connecting structure **400** suitable for removably connecting a display screen **402** to a display device body **404**. In this example, the hook component **406** is thinner than the hook component **212** shown in FIGS. 2 and 3, and comprises a flange **408** configured to provide a suitably large bonding area at which the hook component **406** is adhered to the display screen **402**. Such a structure may allow a display device to have a thinner bezel.

(22) The connecting structure **400** also comprises a compressible structure **410** coupled to one of the body **404** and the hook component **406** at location where the body **404** interfaces the hook component **406**. The compressible structure **410** is configured to compress when the hook component **406** and tab component **412** engage, thereby allowing a firm fit due to opposing force provided by the compressible material. This may help to provide an amount of travel distance between the body **404** and the display screen **402** when the connecting structure is engaged while ensuring a firm fit. Any suitable material(s) may be used as the compressible structure **410**. In some examples, the compressible structure **410** comprises an elastomer or resilient polymer foam. In other examples, the compressible structure may comprise a mechanically compressible structure, such as a leaf spring.

(23) Returning to FIG. 3, the connecting structure **202** comprises an actuation mechanism that is actuatable to selectively engage the tab component **208** and the hook component **212**. Any suitable actuation mechanism may be used. In FIG. 3, the actuation mechanism takes the form of a cam **218** positioned between a first follower **220A** and a second follower **220B** of the tab component. The cam **218** comprises a shaft **225** to which a cam arm **222** is attached, and a pin **224** extending upward from the cam arm **222**. In this example, rotation of the shaft **225** (e.g. via a screwdriver, hex wrench or other suitable tool) causes the pin **224** to apply a force to the first follower **220A** or the second follower **220B**, depending upon the direction of rotation. Force applied to the first follower **220A** or the second follower **220B** causes the tab component **208** to translate relative to the hook component **212**, thereby engaging or disengaging the one or more tabs **210** and the one or more hooks **214**.

(24) The actuation mechanism is accessible from an exterior side of the body **110**. In the example of FIG. 3, the shaft **225** is positioned within a boss **226** formed in the body **110** that extends to an exterior, non-display side of the body **110** (as indicated by dashed lines in FIG. 3) such that a mechanical interface **228** of the actuation mechanism is accessible from the exterior side of the body **110**. The mechanical interface **228** of the actuation mechanism may comprise a hand tool interface configured to be used with common tools (e.g. a screwdriver, a hex wrench, etc.). In other examples, the mechanical interface **228** may comprise a lever, a magnet, or any other suitable structure.

(25) The disclosed connecting structure **202** may allow for the quick and easy attachment of a display screen to a display device body by simply mounting the hook component **212** and tab component **208** to respectively to one of and the other of the device body and the display screen, installing cam **218** into boss **226**, and using the actuation mechanism to engage the hook

component **212** and the tab component **208**.

(26) In some examples, a display device may include two or more connecting structures arranged along different sides of the display screen **108**. Rather than including a separate actuation mechanism along each side of the display screen **108**, such a display device may include a single actuation mechanism configured to actuate both of two or more connecting structures. FIG. 5 schematically shows an example actuation mechanism **500** configured to provide a single point of actuation for two connecting structures. The actuation mechanism **500** comprises a rotatable linkage **502** connecting a first connecting structure **504** arranged along a first side of a display screen **508** to a second connecting structure **506** arranged along a second side of the display screen **508** that is adjacent to the first side.

(27) The rotatable linkage **502** is rotatable to selectively engage the one or more tabs and the one or more hooks of the first connecting structure **504** and to selectively engage the one or more second tabs and the one or more second hooks of the second connecting structure **506**. In this example, the rotatable linkage moves the tab components. In other examples, the rotatable linkage may move the hook components.

(28) While a single point of actuation **512** for two connecting structures **504** and **506** is depicted in FIG. 5, the single point of actuation **512** also may be used to actuate three or more connecting structures. In one specific example, a display device comprises a first connecting structure extending at least a portion of a left side of the display screen **508**, a second connecting structure extending across a bottom side of the display screen **508**, a first rotational linkage connecting the first connecting structure to the second connecting structure, a third connecting structure extending at least a portion of a right side of the display screen **508**, and a second rotational linkage connecting the second connecting structure to the third connecting structure. In such an example, a single actuator may be rotated to engage the hooks and tabs of each of the first, second, and third connecting structures.

(29) Returning briefly to FIG. 3, in some examples, the connecting structure **202** (or connecting structure **400** in FIG. 4) may extend along substantially an entire length of a side of the display screen. However, display devices often have various hardware components that extend through a display device body to implement interfaces. Examples include buttons, cable ports, speakers and microphones. Thus, to accommodate such devices while maintaining a thin device profile, in some examples a display device may include a connecting structure of the type shown in FIGS. 2-4 along a portion of a side of a display screen, and also include a different securing mechanism, with a lesser depth, along a different portion of the same side of the display screen to accommodate such interface hardware.

(30) FIG. 6 schematically a display device **600**, and illustrates portions **602A-602D** of a side of a display screen **608** that may be secured to the body **610** via a connecting structure **202** (FIG. 2) or **400** (FIG. 4) (as examples), and also portions **604A-604C** of the side that may be secured to the body **610** via a different connecting structure to accommodate interface hardware. FIG. 7 depicts a sectional view of the display device **600**, taken along line 7-7 of FIG. 6, to illustrate an example frictional connecting structure **700** that may be used accommodate hardware interface components. The frictional connecting structure **700** comprises a flange **701** that is coupled to the display screen **608** (e.g. via an adhesive), and a protrusion **702** that extends into a complementary slot **704** by a frictional press fit. In FIG. 7, the complementary slot **704** is formed within the body **610**, but also may be implemented in a component attached to the body **610** in other examples. In yet other examples, the frictional connecting structure **700** may be coupled to the body **610** and the protrusion **702** may be held in a complementary slot on the display screen **608** (or a complementary slot of a separate structure coupled to the display screen **608**).

(31) The frictional press fit secures the display screen **608** to the body **610** in a manner that utilizes a relatively small amount of space in a thickness dimension d (e.g. 1-2 mm), and thus provides enough vertical space for a hardware interface component. An example interface hardware

component is shown schematically as a button **612**.

(32) The frictional connecting structure **700** may be formed from any suitable material, and comprise any suitable configuration of feature(s) able to achieve a frictional press fit in the complementary slot. For example, the protrusion **702** may be formed from sheet metal, and include one or more dimples **706** on a surface of the protrusion **702** that provide spring force when inserted in the complementary slot **704**. In other examples, the protrusion may be formed from a polymer that is deformable, and may include surface features (e.g. bumps) that compress and deform when the protrusion is press-fit into the complementary slot **704**. Other examples of display devices may not include such interface hardware extending through the body. In such examples, a connecting structure of a type shown in FIGS. 2-4 may extend substantially along an entire length of a side of the display screen **608**.

(33) FIG. 8 shows an example display device **800** comprising another example connecting structure that may be used along a side of a display device to help retain a display screen **808** within a display device body **810**. The connecting structure of FIG. 8 may be used, for example, where other types of connecting structures (e.g. those of FIGS. 2, 4 and/or 7) are used along other sides. In this example, a fin **801** is coupled to the display screen **808** and inserted behind an interior ledge **802** of the body **810** to hold the display screen **808** to the body **810**. While depicted as having a generally flat profile where the fin **801** extends beneath the ledge **802**, in other examples a fin may have other shapes, such as a hook shape that engages with an interior ledge of the body **810** to provide retention.

(34) The fin **801** may extend along substantially an entire length of a side of the display screen **808**, or may extend a portion of a side of the display screen **808**. Further, in some examples, multiple fins may be provided along a length of a side of a display screen. During installation of the display screen **808**, the display screen **808** may be inserted at an angle to position the fin **801** beneath the interior ledge **802** of the body **810**, and then set into the body **810** and secured via other connecting structures along other sides.

(35) FIG. 9 is a flowchart illustrating an example method **900** for securing a display screen to a body of a device. In various examples, method **900** may be implemented as a manufacturing process or as a repair process.

(36) At **902**, method **900** comprises coupling a tab component of a connecting structure to one of the display screen and the body. In some examples, the tab component may be a first tab component, and method **900** optionally may comprise coupling a second tab component of a second connecting structure to a one of the display screen and the body along a different side than the first tab component, as indicated at **904**. As described above, the tab component comprises one or more tabs, and the second tab component comprises one or more second tabs.

(37) At **906**, method **900** comprises coupling a hook component of the connecting structure to the other of the display screen and the body. In some examples, the hook component may be a first hook component, and method **900** optionally may comprise coupling a second hook component of the second connecting structure to the other of the display screen and the body, as indicated at **908**. As described above, the hook component comprises one or more hooks complementary to the one or more tabs, and the second hook component comprises one or more second hooks complementary to the one or more second tabs.

(38) Method **900** optionally may comprise attaching other securing mechanism(s) to one or more of the display screen and the body, as indicated at **910**. In some examples, attaching other securing mechanism(s) may comprise coupling a fin to a different side of the display screen than the connecting structure, as indicated at **912**. Further, in some examples, a display device may include a different securing mechanism along a different portion of a same side of the display screen as the connecting structures of processes **902** and **906**, e.g. to accommodate a hardware interface component that extends through a display device body. Thus, attaching other securing mechanism(s) may comprise coupling a frictional connecting structure to one of the body and the

display screen, as indicated at **914**.

(39) At **916**, method **900** comprises inserting the display screen into the body. Inserting the display screen into the body may comprise inserting the fin behind an interior ledge of the body to hold the display screen to the body, as indicated at **918**. Inserting the display screen into the body also may comprise inserting a protrusion of the frictional connecting structure into a complementary slot such that the protrusion is held in the complementary slot by a frictional press fit, as indicated at **920**.

(40) At **922**, method **900** comprises using the actuation mechanism to engage the tab component and the hook component. As described above, the actuation mechanism is accessible from an exterior side of the body. In some examples, the actuation mechanism comprises a cam having a rotatable shaft coupled to a cam arm and a pin extending upwards from the cam arm. Rotating the rotatable shaft causes the pin to push against one of the hook component and the tab component, and thereby move the tab component and the hook component into or out of engagement depending upon the direction of rotation. In other examples, the actuation mechanism comprises a rotatable linkage connecting the first connecting structure to the second connecting structure. Actuating the actuation mechanism may comprise rotating the rotatable linkage to engage the tab component and hook component of the first connecting structure and to engage the second tab component and second hook component of the second connecting structure, as indicated at **924**. In yet other examples, any other suitable actuation mechanism may be used.

(41) While described herein in the context of display devices, the disclosed examples may alternatively be used in other contexts. For example, the disclosed examples may be used to selectively secure a door or access panel to a machine body.

(42) Another example provides a display device, comprising a body, a display screen, and a connecting structure comprising a tab component having one or more tabs, the tab component being coupled to one of the display screen and the body, a hook component comprising one or more hooks complementary to the one or more tabs, the hook component coupled to the other of the display screen and the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and an actuation mechanism accessible from an exterior side of the body, the actuation mechanism being actuatable to engage the tab component and the hook component. In such an example, the hook component may additionally or alternatively be adhered to the display screen. In such an example, a portion of the tab component may additionally or alternatively be positioned within a slot formed in the body. In such an example, the actuation mechanism may additionally or alternatively comprise a cam positioned between a first follower and a second follower of the first connecting structure. In such an example, the connecting structure may additionally or alternatively be a first connecting structure arranged along a first side of the display screen, the display device may additionally or alternatively comprise a second connecting structure arranged along a second side of the display screen, the second connecting structure may additionally or alternatively comprise a second tab component having one or more second tabs, the second tab component being coupled to the one of the display screen and the body, and a second hook component comprising one or more second hooks complementary to the one or more second tabs, the second hook component coupled to the other of the display screen and the body, the second tab component and the second hook component being moveable relative to one another to selectively engage the one or more second tabs with the one or more second hooks. In such an example, the actuation mechanism may additionally or alternatively comprise a rotatable linkage connecting the first connecting structure to the second connecting structure, the rotatable linkage being rotatable to engage the tab component and the hook component of the first connecting structure and to engage the second tab component and the second hook component of the second connecting structure. In such an example, the display device may additionally or alternatively comprise a fin arranged along a different side of the display screen than the connecting structure, the fin inserted behind an interior

ledge of the body to hold the display screen to the body. In such an example, the fin may additionally or alternatively be adhered to the display screen. In such an example, the connecting structure may additionally or alternatively be arranged along a first portion of a side of the display screen, the display device may additionally or alternatively comprise a frictional connecting structure extending along a second portion of the side of the display screen, the frictional connecting structure coupled to one of the body and the display screen, the frictional connecting structure comprising a protrusion that is held in a complementary slot on the other of the body and the display screen by a frictional press fit. In such an example, the connecting structure may additionally or alternatively comprise a metal.

(43) Another example provides a method for securing a display screen to a body of a device, the method comprising coupling a tab component of a connecting structure to one of the display screen and the body, the tab component having one or more tabs, coupling a hook component of the connecting structure to the other of the display screen and the body, the hook component having one or more hooks complementary to the one or more tabs, inserting the display screen into the body such that the tab component and the hook component are moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and actuating an actuation mechanism to engage the tab component and the hook component, the actuation mechanism being accessible from an exterior side of the body. In such an example, the connecting structure may additionally or alternatively be a first connecting structure, the method may additionally or alternatively comprise, prior to inserting the display screen into the body, coupling a second tab component of a second connecting structure to the one of the display screen and the body along a second side of the display screen, the second tab component having one or more second tabs, and coupling a second hook component of the second connecting structure to the other of the display screen and the body along the second side of the display screen, the second hook component having one or more second hooks complementary to the one or more second tabs. In such an example, the actuation mechanism may additionally or alternatively comprise a rotatable linkage connecting the first connecting structure to the second connecting structure, and actuating the actuation mechanism may additionally or alternatively comprise rotating the rotatable linkage to engage the tab component and the hook component of the first connecting structure and to engage the second tab component and the second hook component of the second connecting structure. In such an example, the method may additionally or alternatively comprise, prior to inserting the display screen into the body, coupling a fin to a different side of the display screen than the connecting structure, and inserting the display screen into the body may additionally or alternatively comprise inserting the fin behind an interior ledge of the body to hold the display screen to the body. In such an example, the connecting structure may additionally or alternatively be arranged along a first portion of the side of the display screen, the method may additionally or alternatively comprise, prior to inserting the display screen into the body, coupling a frictional connecting structure to one of the body and the display screen, the frictional connecting structure extending along a second portion of the side of the display screen, and inserting the display screen into the body may additionally or alternatively comprise inserting a protrusion of the frictional connecting structure into a complementary slot such that the protrusion is held in the complementary slot by a frictional press fit.

(44) Another example provides a connecting structure for removably attaching a display screen of a display device to a body of the display device, the connecting structure comprising a tab component having one or more tabs, the tab component configured to be coupled to one of the display screen and the body, a hook component comprising one or more hooks complementary to the one or more tabs, the hook component configured to be coupled to the other of the display screen and the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and an actuation mechanism actuatable to engage the tab component and the hook component. In such an example,

the hook component may additionally or alternatively be configured to be adhered to the display screen, and a portion of the tab component may additionally or alternatively be configured to be inserted within a slot formed in the body. In such an example, the actuation mechanism may additionally or alternatively comprise a hand tool interface on an end of a shaft. In such an example, the shaft may additionally or alternatively be coupled to a cam arm, the cam arm may additionally or alternatively comprise a pin configured to be positioned between a first follower and a second follower of the tab component. In such an example, the connecting structure may additionally or alternatively comprise a second tab component having one or more second tabs, the second tab component configured to be coupled to the one of the display screen and the body along a different side of the display screen than the tab component, and a second hook component comprising one or more second hooks complementary to the one or more second tabs, the second hook component configured to be coupled to the other of the display screen and the body along the different side of the display screen, and the actuation mechanism may additionally or alternatively comprise a rotatable linkage connecting the first tab component to the second tab component, the rotatable linkage being actuatable to engage the tab component and the hook component and to engage the second tab component and the second hook component.

(45) It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.

(46) The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A display device, comprising: a body; a display screen; and a connecting structure comprising a tab component having one or more tabs, the tab component being coupled to one of the display screen or the body, wherein a portion of the tab component is moveably positioned within a slot formed in one of the display screen or the body, a hook component comprising one or more hooks complementary to the one or more tabs, the hook component fixedly coupled to the other of the display screen or the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks, and an actuation mechanism accessible from an exterior side of the body, the actuation mechanism being actuatable to engage the tab component and the hook component.
2. The display device of claim 1, wherein the hook component is adhered to the other of the display screen or the body.
3. The display device of claim 1, wherein the actuation mechanism comprises a cam positioned between a first follower and a second follower of the first connecting structure.
4. The display device of claim 1, wherein the connecting structure is a first connecting structure and is arranged along a first side of the display screen, the display device further comprising a second connecting structure arranged along a second side of the display screen, the second connecting structure comprising a second tab component having one or more second tabs, the second tab component being coupled to the one of the display screen or the body, and a second hook component comprising one or more second hooks complementary to the one or more second tabs, the second hook component coupled to the other of the display screen or the body, the second tab component and the second hook component being moveable relative to one another to selectively engage the one or more second tabs with the one or more second hooks.

5. The display device of claim 4, wherein the actuation mechanism further comprises a rotatable linkage connecting the first connecting structure to the second connecting structure, the rotatable linkage being rotatable to engage the tab component and the hook component of the first connecting structure and to engage the second tab component and the second hook component of the second connecting structure.
6. The display device of claim 1, further comprising a fin arranged along a different side of the display screen than the connecting structure, the fin inserted behind an interior ledge of the body to hold the display screen to the body.
7. The display device of claim 6, wherein the fin is adhered to the display screen.
8. The display device of claim 1, wherein the connecting structure is arranged along a first portion of a side of the display screen, the display device further comprising: a frictional connecting structure extending along a second portion of the side of the display screen, the frictional connecting structure coupled to one of the body or the display screen, the frictional connecting structure comprising a protrusion that is held in a complementary slot on the other of the body or the display screen by a frictional press fit.
9. The display device of claim 1, wherein the connecting structure comprises a metal.
10. A method for securing a display screen to a body of a device, the method comprising: coupling a tab component of a connecting structure to one of the display screen or the body, wherein a portion of the tab component is moveably positioned within a slot formed in the one of the display screen or the body, the tab component having one or more tabs, coupling a hook component of the connecting structure to the other of the display screen or the body, the hook component having one or more hooks complementary to the one or more tabs; inserting the display screen into the body such that the tab component and the hook component are moveable relative to one another to selectively engage the one or more tabs with the one or more hooks; and actuating an actuation mechanism to engage the tab component and the hook component, the actuation mechanism being accessible from an exterior side of the body.
11. The method of claim 10, wherein the connecting structure is a first connecting structure, the method further comprising, prior to inserting the display screen into the body: coupling a second tab component of a second connecting structure to one of the display screen or the body along a second side of the display screen, the second tab component having one or more second tabs, and coupling a second hook component of the second connecting structure to the other of the display screen or the body along the second side of the display screen, the second hook component having one or more second hooks complementary to the one or more second tabs.
12. The method of claim 11, wherein the actuation mechanism comprises a rotatable linkage connecting the first connecting structure to the second connecting structure, and wherein actuating the actuation mechanism comprises rotating the rotatable linkage to engage the tab component and the hook component of the first connecting structure and to engage the second tab component and the second hook component of the second connecting structure.
13. The method of claim 10, further comprising: prior to inserting the display screen into the body, coupling a fin to a different side of the display screen than the connecting structure; and wherein inserting the display screen into the body comprises inserting the fin behind an interior ledge of the body to hold the display screen to the body.
14. The method of claim 10, wherein the connecting structure is arranged along a first portion of the side of the display screen, the method further comprising, prior to inserting the display screen into the body, coupling a frictional connecting structure to one of the body or the display screen, the frictional connecting structure extending along a second portion of the side of the display screen; and wherein inserting the display screen into the body comprises inserting a protrusion of the frictional connecting structure into a complementary slot such that the protrusion is held in the complementary slot by a frictional press fit.
15. A connecting structure for removably attaching a display screen of a display device to a body of

the display device, the connecting structure comprising: a tab component having one or more tabs, the tab component configured to be coupled to one of the display screen or the body; a hook component comprising one or more hooks complementary to the one or more tabs, the hook component configured to be fixedly coupled to the other of the display screen or the body, the tab component and the hook component being moveable relative to one another to selectively engage the one or more tabs with the one or more hooks; and an actuation mechanism actuatable to engage the tab component and the hook component.

16. The connecting structure of claim 15, wherein the hook component is configured to be adhered to the display screen.

17. The connecting structure of claim 15, wherein the actuation mechanism comprises a hand tool interface on an end of a shaft.

18. The connecting structure of claim 17, wherein the shaft is coupled to a cam arm, the cam arm comprising a pin configured to be positioned between a first follower and a second follower of the tab component.

19. The connecting structure of claim 15, further comprising: a second tab component having one or more second tabs, the second tab component configured to be coupled to one of the display screen or the body along a different side of the display screen than the first tab component; and a second hook component comprising one or more second hooks complementary to the one or more second tabs, the second hook component configured to be coupled to the other of the display screen or the body along the different side of the display screen; and wherein the actuation mechanism comprises a rotatable linkage connecting the first tab component to the second tab component, the rotatable linkage being actuatable to engage the tab component and the hook component and to engage the second tab component and the second hook component.

20. The connecting structure of claim 15, wherein the tab component is configured to be adhered to the display screen.
